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MILITARY SPECIFICATION

MECHANISM, RECOIL, 105-MM HOWITZER, M2A5, W/REPAIR PARTS AND W/O EQUIPMENT OR ACCESSORIES

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of hydro-pneumatic recoil mechanism designed for use with the 105-mm howitzer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-6872 — Soldering Process: General Specification for.

MIL-W-13748 — Welding, Metal Arc: Steels, Constructional Readily Weldable for Highly Stressed Joints.

MIL-P-14232 — Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL-W-14539 — Weapon Parts for Antiaircraft Artillery, Rocket Launchers and Small Arms, Packaging of.

MIL-I-45208 — Inspection Requirements, General Specification for.

STANDARDS

MILITARY

MIL-STD-8 — Dimensioning and Tolerancing.

MIL STD-10 — Surface Roughness, Waviness and Lay.

JAN-STD-19 — Welding Symbols.

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130 — Identification Marking of U.S. Military Property.

MIL-STD-171 — Finishing of Metal and Wood Surfaces.

FSC 1015

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DRAWINGS

ORDNANCE CORPS

F8412320 — Mechanism, Recoil, 105-mm Howitzer, M2A5.

PUBLICATIONS

ORDNANCE CORPS

Packaging Data Sheet — MIL–W–14539–P8412320.

LSQAP–8412320—Index of Supplementary Quality Assurance Provisions.

IEL–8412320 — Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with material specification referenced on applicable drawings.

3.2 **Dimensions.** Dimensions of parts and assemblies shall be in accordance with Drawing F8412320 and the detail drawings listed thereon. Interpretation of dimensional requirements shall be in accordance with Standard MIL–STD–8.

3.3 **Construction.** Parts and assemblies shall be positioned and attached in accordance with Drawing F8412320.

3.3.1 *Welding.* Welding of parts and assemblies shall be in accordance with Specification MIL–W–13748. Interpretation of weld symbols shall be as defined in Standard JAN–STD–19.

3.3.2 *Soldering.* Soldering of parts shall be in accordance with Specification MIL–S–6872.

3.3.3 *Surface finish.* Surface finish shall be as specified on applicable drawings. Interpretation of surface finish shall be as defined in Standard MIL–STD–10.

3.4 **Performance and product characteristics.**

3.4.1 *Recoil mechanism.* Each completed recoil mechanism shall conform to the general characteristics specified on Drawing F8412320 and as specified herein.

3.4.2 *Friction.* Packings shall be carefully adjusted so that the recoil mechanism shall function with a minimum of friction. Friction of the floating piston and the recoil piston rod and stuffing box shall be within the limits specified in table I, after assembly, after gymnastication and after proof-testing.

TABLE I. *Friction in pounds per square inch of hydraulic pressure after proof-testing*

Part name	Minimum	Maximum
Floating piston	50	250
Recoil piston rod and stuffing box	50	280

3.4.3 *Air filling valve.* The air filling valve shall show no evidence of nitrogen leakage when tested in accordance with 4.4.3.

3.4.4 *Regulator valves.* The regulator valves shall show no evidence of oil leakage when tested in accordance with 4.4.4.

3.4.5 *Oil index.* The oil index shall function without evidence of interference throughout its entire range of travel. There shall be only sufficient oil leakage to provide the necessary lubrication.

3.4.6 *Respirator.* The respirator shall control the velocity of the counter-recoil action by slowing up the return of the howitzer to in-battery position without deformation to the recoiling parts.

3.4.7 *Gymnastication.* After final assembly, prior to proof-testing each recoil mechanism shall be gymnasticated as specified in 4.4.6 for

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conformance to performance requirements referenced in this specification.

3.4.8 *Proof-testing.* Each recoil mechanism shall be fully assembled and lubricated for proof-testing. Proof-testing shall be in accordance with table III. There shall be no evidence of malfunction, deformation of parts, erratic motion of recoil and counter-recoil or oil leakage other than specified in 3.4.5 as a result of this test.

3.5 Painting. Unless otherwise specified, recoil mechanism shall be painted in accordance with finish No. 5.3.2.3 and system No. 21.5 of Standard MIL–STD–171. Parts not to be painted shall be in accordance with the same standard. Color shall be specified by the procuring activity (see 6.2).

3.6 Marking. Markings, including identification and instructional shall be in accordance with applicable drawings and Standard MIL–STD–130 (see 6.2).

3.7 Hydraulic fluid. Hydraulic fluid shall be in accordance with the Specification referenced on Drawing F8412320.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities or services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Contractor's inspection system.* The contractor shall provide and maintain an adequate inspection system in accordance with Specification MIL–I–45208.

4.2 Inspection provisions.

4.2.1 *Materials.* Materials shall be tested in accordance with the applicable material specification referenced on the drawings.

4.2.2 *Welding.* Inspection and tests of welds shall be in accordance with Specification MIL–W–13748. Actual production of welds shall not proceed prior to Government approval of the contractor's welding procedure and test specimens. Radiograph inspection of welds shall be in the locations specified on the applicable drawings.

4.2.3 *Protective finishes.* Inspection and tests of protective finishes and painting shall be in accordance with Standard MIL–STD–171.

4.2.4 *Commercial parts.* The Government shall be furnished by the contractor a statement of findings. This statement of findings shall list the inspections performed and the results obtained and attest that the specified Air Force-Navy (AN), Military (MS), and Ordnance Engineering Standard parts are in accordance with the applicable standard drawings.

4.2.5 *Sampling.*

4.2.5.1 *Lot formation.* A lot shall consist of all recoil mechanisms made under essentially like conditions and offered for delivery at any one time.

4.2.5.2 *Complete recoil mechanism.* Sampling for examination shall be in accordance with level II of Standard MIL–STD–105.

4.3 Examination.

4.3.1 *Parts and assemblies.* Examination of parts and assemblies shall be in accordance with quality assurance documents listed in the Index of Supplementary Quality Assurance Provisions LSQAP–8412320.

4.3.2 *Recoil mechanism.* Each of the samples selected in accordance with 4.2.5.2 shall

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be examined to verify conformance with this specification. Examination shall be conducted in accordance with table II. Any recoil mechanism in the sample containing one or more defects shall be rejected and if the number of defective mechanisms in any sample exceeds the acceptance number of that sample, the lot represented by the sample shall be rejected.

TABLE II. *Classification of defects in accordance with Standard MIL–STD–105*

Categories	Defects	Inspection method
Critical	None defined.	
Major	AQL = 1.5 percent:	
101	Missing or improper parts	Visual
102	Improperly attached parts	Visual
Minor	AQL = 4.0 percent:	
201	Identification marking missing or incorrect.	Visual
202	Visible defects of painting including incomplete coverage, runs, sags, blisters, and incorrect color.	Visual

4.3.3 *Packaging and packing.* Examination of packaging and packing shall be in accordance with Specification MIL–P–14232.

4.4 **Test procedures.**

4.4.1 *Parts and assemblies.* Testing of parts and assemblies including hydrostatic test of cylinders and pull test of rods shall be in accordance with the quality assurance documents listed in the Index of Supplementary Quality Assurance Provisions, LSQAP–8412320 and as specified herein.

4.4.2 *Friction.* Manometer readings shall be taken on each recoil mechanism after assembly, after resting 48 hours, after gymnastication, and after proof-testing to determine conformance to 3.4.2.

4.4.3 *Air filling valve.* Each recuperator cylinder head assembly shall be mounted in a suitable test fixture and subjected to 1,000 pounds per square inch nitrogen pressure, and examined to determine conformance to 3.4.3.

4.4.4 *Regulator valves.* Each regulator valve shall be tested by applying 20 pounds per square inch nitrogen pressure against the valve head to determine conformance to 3.4.4.

4.4.5 *Oil index.* Immediately after gymnastication, the reserve oil shall be drawn off and examined for a foamy or emulsified condition. A foamy or emulsified condition of the oil shall be cause for rejection of the mechanism. After the reserve oil has been replaced the oil index shall be examined to determine conformance to 3.4.5.

4.4.6 *Gymnastication.* Each recoil mechanism shall be gymnasticated mechanically in recoil and counter-recoil for 75 to 100 cycles to determine conformance to 3.4.7. Recoiling parts shall be retracted from battery 38 inches plus or minus 1 inch and released. Counter-recoil time shall be 2.5 seconds plus or minus .5 seconds. Gymnastication shall be observed by the Government inspector. Failure of the mechanism to perform as intended shall be cause for rejection.

4.4.7 *Proof-testing.* Each recoil mechanism shall be proof-tested by the Government to determine conformance to 3.4.8. Proof-testing shall be conducted in accordance with table III. Any recoil mechanism failing the proof test shall be rejected.

4.4.7.1 *Standard round.* A standard round is one employing a standard weight projectile of 33 pounds and loaded to give a muzzle velocity of 1550 feet per second and having a rated maximum pressure of 32,500 pounds per square inch.

4.4.7.2 *Propellant.* Propellant used in the rounds prescribed in table III shall be from a lot that gives the standard weight projectile of 33 pounds a muzzle velocity of 1,550 feet per second with an average maximum powder pressure of not less than 27,000 pounds per square inch nor more than 32,500 pounds per square inch.

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TABLE III. *Proof-testing*

No. of rounds	Percent of rated maximum pressure (MRP)	Elevation
1	100 percent MRP	0°
1	115 percent MRP	65° (max)
1	100 percent MRP	65° (max)
1	100 percent MRP	0°

(a) Proof-testing for the acceptance of the recoil mechanism may be combined with that of the howitzer and carriage. In such combined tests, rounds in addition to those required by table III may be fired (see 6.2.3).

(b) At the conclusion of the proof-testing, each recoil mechanism shall be carefully examined for oil leaks, breaks or deformations. Manometer tests shall be made and pressures and frictions calculated. Oil leakage from recoil mechanism shall not be more than 10 drops in a 3-minute period and shall be dark in color.

4.4.8 *Inspection equipment.* List of Government inspection equipment of examination and test of parts for the recoil mechanism is contained in Index of Inspection Equipment List, IEL–8412320.

5. PREPARATION FOR DELIVERY

5.1 Recoil mechanism. Preservation, packaging, packing, and marking shall be in accordance with Packaging Data Sheet MIL–W–14539–P8412320.

5.2 Repair parts. Preservation, packaging, packing, and marking shall be in accordance with applicable packaging data sheets of Specification MIL–W–14539.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Title, number, and date of applicable drawings and packaging data sheets.

(c) Place of inspection and tests.

(d) Selection applicable levels of packaging and packing.

6.2 Suggested contractual features.

6.2.1 *Serial number.* The contractor should be furnished a block of serial numbers by the contracting officer for marking of nameplate (see 3.6).

6.2.2 Color Chip No. X24087 (Olive-drab, Semigloss) of Federal Standard 595 shall be furnished the contractor to be used as a guide for painting (see 3.5).

6.2.3 *Proof-testing.* Whenever possible, the recoil mechanism should be proof-tested in combination with unproofed carriages and cannon. When this is done the following firing schedule should be used.

No. of rounds	Percent of rated maximum pressure (MRP)	Elevation	Traverse
1	100 percent MRP	0°	Center
1	115 percent MRP	65° (max)	Center
1	100 percent MRP	65° (max)	(1)
1	100 percent MRP	0°	(1)

(1) These rounds should be fired alternately over the right and left trails.

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Marine Corps

Preparing activity:
Army—Ordnance Corps

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